

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (20 points)*Submit the postlab to the TA at the end of the lab.***The Projector (6 points)****Question 3****1 point**

Add measurements from your study of the projector to Data Table 1 and compute all derived quantities. To estimate the uncertainty on $1/i$, use propagation of uncertainties:

$$\Delta(1/i) = \frac{\Delta i}{i^2}.$$

Data table 1: *Data for the projector measurements.*

Object distance p [cm]	Image distance i [cm]	Image distance uncertainty Δi [cm]	Magnified / Reduced Real / Virtual Inverted / Upright	$1/p$ [cm ⁻¹]	$1/i$ [cm ⁻¹]	$\Delta(1/i)$ [cm ⁻¹]

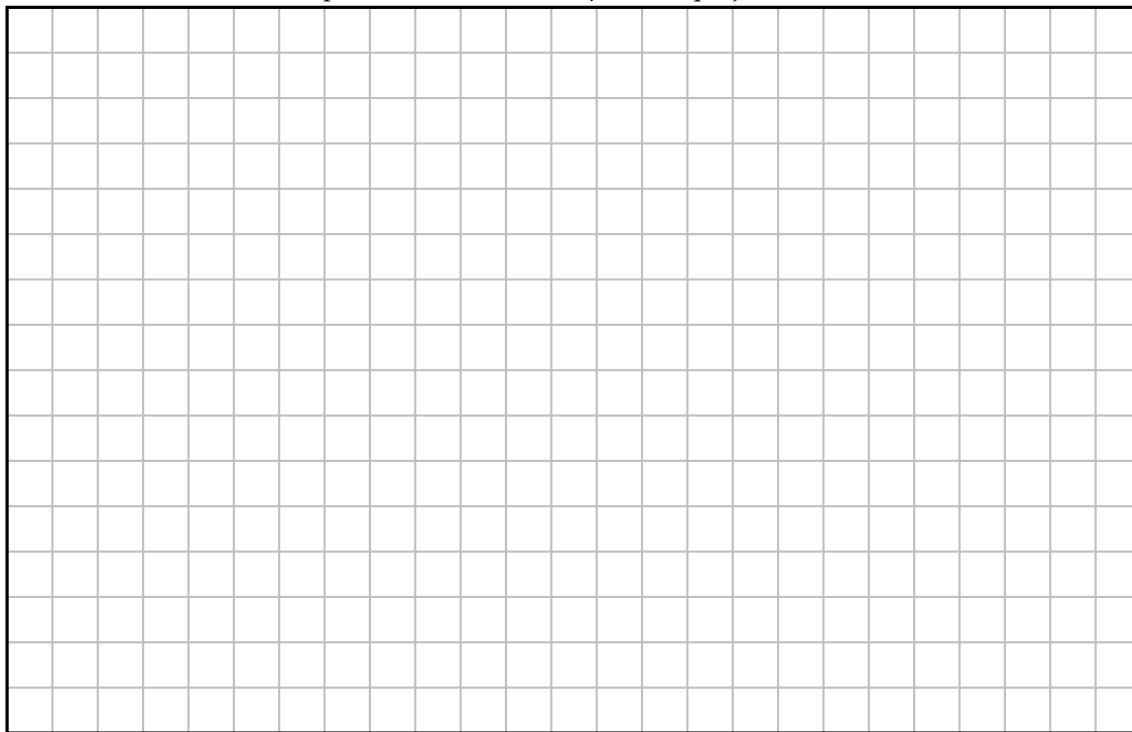
Question 4**1 point**

As the object distance p gets larger, the image distance i approaches some value. Estimate this value by increasing the object distance, and observe the image distance.

Question 5**2 points**

In Graph 1, plot $1/i$ versus $1/p$. Draw the calculated uncertainties on $1/i$ as error bars. Include labels and units for the axes. Draw a best-fit straight line through your data points. Determine the slope and y -intercept of this line. Show your work below.

Graph 1: Plot of $1/i$ vs. $1/p$ for the projector lens.

**Question 6****1 point**

According to the plot in Graph 1, what is the focal length of the lens used in the projector? Use the thin lens formula in eq. (1) and the equation for a line ($y = mx + b$).

Question 7**1 point**

Is the thin lens formula verified according to your data and Graph 1? Explain your answer.

The Magnifier (4 points)**Question 8****1 point**

Add measurements from your study of the magnifier to Data Table 2.

Data table 2: *Data for the magnifier measurements.*

75 mm Lens

height with lens: _____

height without lens: _____

 $M_{75 \text{ mm}}$ (measured): _____**150 mm Lens**

height with lens: _____

height without lens: _____

 $M_{150 \text{ mm}}$ (measured): _____**Question 9****1 point**

Calculate the theoretical magnification of the 75 mm lens and the 150 mm lens using your near point. Show your work.

Question 10**2 points**

How does your measured magnification compare to the theoretical magnification? Explain why it may or may not be so. Give specific reasons having to do with your measurement procedure.

The Compound Microscope (4 points)**Question 11****1 point**

Add measurements from your study of the compound microscope to Data Table 3.

Data table 3: *Data for the compound microscope.*

f_{obj} [cm]:	_____	height with lenses:	_____
f_{eye} [cm]:	_____	height without lenses:	_____
Barrel length L [cm]:	_____	M (measured):	_____

Question 12**1 point**

Calculate the theoretical magnification of your microscope using eq. (6). Show all work.

Question 13**1 point**

Is your calculated magnification consistent with the measured value? Why or why not?

Question 14**1 point**

Why does the magnification increase as the objective lens is moved closer to the object? What focusing problems tend to develop as the magnification increases?

The Astronomical Telescope (3 points)**Question 15****1 point**

Add measurements from your study of the astronomical telescope to Data Table 4.

Data table 4: *Data for the astronomical telescope.*

f_{obj} [cm]:	_____	height with lenses:	_____
f_{eye} [cm]:	_____	height without lenses:	_____
Upright or inverted?	_____	M (measured):	_____

Question 16**1 point**

Calculate the theoretical magnification of your astronomical telescope in eq. (7). Show your work.

Question 17**1 point**

Compare your calculated magnification to the measured one. Compute the percentage difference using the expression

$$\left| \frac{M_{\text{meas}} - M_{\text{calc}}}{M_{\text{calc}}} \right| \times 100 [\%].$$

Is the percentage difference under 20%? If not, explain why not. Are the two values consistent with each other? Justify your answer.

The Terrestrial Telescope (3 points)**Question 18****1 point**

Add measurements from your study of the terrestrial telescope to Data Table 5.

Data table 5: *Data for the terrestrial telescope.*

f_{obj} [cm]:	_____	height with lenses:	_____
f_{eye} [cm]:	_____	height without lenses:	_____
Upright or inverted?	_____	M (measured):	_____

Question 19**1 point**

Calculate the theoretical magnification of your terrestrial telescope in eq. (7). Show your work.

Question 20**1 point**

What is the main difference between the two telescopes' performance (other than the eyepiece)? When would you use the astronomical telescope? When would you use the terrestrial telescope?